

RESERVE BANK OF INDIA: MASTER DIRECTION AND GUIDELINES ON MODEL RISK MAN

CHAPTER I: REGULATORY FRAMEWORK

1. Model Risk Definition and Scope.

Model risk is defined as the potential for adverse consequences resulting from decisions based on incorrect or misused model outputs and reports.

All commercial banks, non-banking financial companies (NBFCs), and all-India financial institutions shall establish a comprehensive Model Risk Management (MRM) framework.

CHAPTER II: MODEL DEVELOPMENT AND VALIDATION

Section 4: Sound Model Development Practices.

- (1) Financial institutions must rigorously document model design, theoretical rationale, mathematical logic, assumptions, data sources, and sample selection.
- (2) Machine learning and quantitative trading models must undergo robust feature importance analysis, sensitivity testing, and out-of-time validation before deployment.

Section 5: Independent Model Validation.

- (1) All credit scoring, market risk, and algorithmic models must be independently validated by a qualified team separate from model development.
- (2) Validation shall evaluate conceptual soundness, ongoing monitoring, outcome analysis, benchmarking against alternative models, and backtesting against historical market regimes.

CHAPTER III: GOVERNANCE AND AUDIT TRAILS

Section 7: Model Inventory and Annual Review.

- (1) Regulated entities shall maintain an enterprise-wide model inventory cataloging all active, retired, and under-development models.
- (2) High-risk models (including capital calculation and automated algorithmic decision systems) shall be reviewed and re-validated at least annually.
- (3) Complete audit trails of code versioning, parameter updates, training datasets, and performance degradation alerts must be preserved for at least five (5) years.